

❖ Android XML UI Elements

1. TextView:-

TextView is used to display text on the Android screen.

Tag:- <TextView> </TextView>

Main Attributes:-

Attribute	Purpose
android:id	Gives unique ID
android:text	Displays text
android:textSize	Sets text size
android:textColor	Sets text color
android:textStyle	Bold/italic text
android:gravity	Positions text
android:fontFamily	Sets font
android:padding	Adds inner space

Example

```
<TextView  
    android:id="@+id/txtWelcome"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Welcome to Android"  
    android:textSize="24sp"  
    android:textColor="#000000"  
    android:textStyle="bold" />
```

2. EditText:-

EditText is used to take input from the user.

Examples:

- Name
- Email
- Password
- Mobile number

Tag:- <EditText></EditText>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:hint	Shows instruction inside field
android:text	Sets default text
android:inputType	Defines type of input
android:maxLength	Limits characters
android:textSize	Sets text size
android:textColor	Sets text color

Example:-

```
<EditText  
    android:id="@+id/edtName"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:hint="Enter your name"  
    android:inputType="text" />
```

Password Example

```
<EditText
    android:id="@+id/edtPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter password"
    android:inputType="textPassword" />
```

3. Button:-

Button is used to allow the user to perform an action by clicking.

Examples:-

- Login
- Register
- Submit
- Save
- Delete

Tag:- <Button></Button>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:text	Button text
android:enabled	Enables/disables button
android:visibility	Controls visibility
android:padding	Inner spacing
android:layout_margin	Outer spacing

Example

```
<Button  
    android:id="@+id/btnLogin"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Login" />
```

4. ImageView:-

ImageView is used to display an image.

Examples:-

- Logo
- Profile photo
- Product image
- Icons

Tag:- <ImageView></ImageView>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:src	Image source
android:scaleType	Controls image scaling
android:contentDescription	Describes image
android:background	Sets background

Example:-

```
<ImageView  
    android:id="@+id/imgLogo"  
    android:layout_width="150dp"  
    android:layout_height="150dp"  
    android:src="@drawable/logo"  
    android:scaleType="centerCrop"  
    android:contentDescription="App Logo" />
```

5. CheckBox:-

- CheckBox allows the user to select or unselect an option.
- Multiple checkboxes can be selected at the same time.

Tag:- <CheckBox></CheckBox>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:text	Checkbox text
android:checked	Default checked state
android:enabled	Enable/disable
android:buttonTint	Changes checkbox color

Example:-

```
<CheckBox  
    android:id="@+id/chkRemember"
```

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Remember Me"
android:checked="false" />
```

6. RadioButton:-

- RadioButton is used when the user needs to select one option from multiple options.
- Usually used inside RadioGroup.

Tag:- <RadioButton></RadioButton>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:text	Option text
android:checked	Default selection
android:enabled	Enable/disable

Example:-

```
<RadioButton
    android:id="@+id/radioMale"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Male" />
```

7. RadioGroup

-RadioGroup groups multiple RadioButtons together so that only one option can be selected.

Tag:- <RadioGroup></RadioGroup>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:orientation	Horizontal/vertical arrangement
android:checkedButton	Selects a default RadioButton

Example:-

```
<RadioGroup
    android:id="@+id/radioGroupGender"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical">
    <RadioButton
        android:id="@+id/radioMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male" />

    <RadioButton
        android:id="@+id/radioFemale"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female" />
</RadioGroup>
```

8 . ToggleButton:-

ToggleButton provides a button with two states, such as ON/OFF.

Tag:- <ToggleButton></ToggleButton>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:textOn	Text when ON
android:textOff	Text when OFF
android:checked	Initial state

Example:-

```
<ToggleButton
    android:id="@+id/toggleStatus"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textOn="ON"
    android:textOff="OFF"
    android:checked="false" />
```

9. Spinner:-

Spinner displays a dropdown list from which the user can select an option.

Tag:- <Spinner>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:spinnerMode	Dropdown/dialog style
android:entries	Provides predefined items

Example:-

```
<Spinner
    android:id="@+id/spinnerCity"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:entries="@array/cities" />
```

Possible output:

Select City ▼

Pune

Mumbai

Nashik

Sangli

10. ProgressBar:-

ProgressBar shows that a process is running.

Examples:

- File downloading
- Data loading
- API request
- Login processing

Tag:- <ProgressBar>

Main Attributes:-

Attribute	Purpose
android:id	Unique ID
android:progress	Current progress
android:max	Maximum value
android:indeterminate	Unknown/continuous progress
android:visibility	Shows/hides it

Example:-

```
<ProgressBar  
    android:id="@+id/progressBar"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:indeterminate="true" />
```